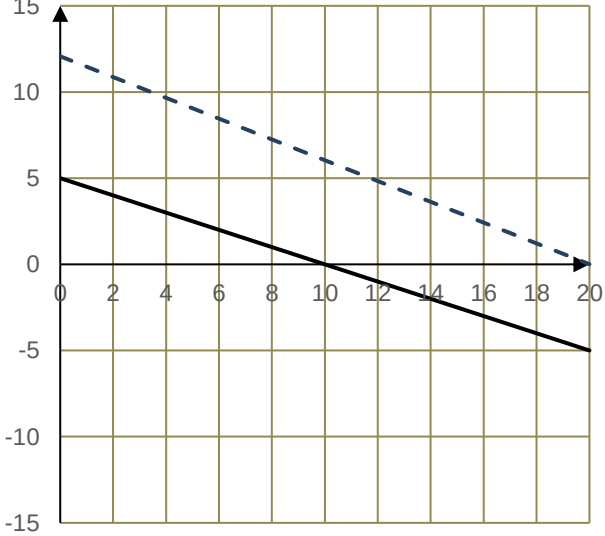


Questions			Answers	Marks
2	(a)		apply equation of motion $v = u + at$ vertically, we get $-5 = +5 - 9.81t$ for substitution	M1
			$t = 1.019 \text{ s}$ for intermediate value	A1
	(b)		$\frac{20}{1.019} = 19.62 \text{ m s}^{-1}$	B1
			Alternative, Using $v_y = u_0 + at$ and substituting $t = \frac{x}{v_x}$, we get $v_y = u_0 - g \frac{1}{v_x} x$. So gradient $= -\frac{1}{2} = -\frac{9.81}{v_x}$	

	(c)		$\tan \theta = \frac{5.0}{19.62}$	M1
			$\theta = 14.3^\circ$	A1
	(d)		x-intercept at (20,0) (because at maximum height $v_y = 0$) y-intercept above 5 (because with $\theta < 45^\circ$ to increase range , <u>θ needs to increase</u>, and $v \sin \theta$ needs to increase.) Example:  Extra note As θ increase, and $v_x = v \cos \theta$ decreases. So gradient increases.	B1

3	(c)		The moment of a force is equal to the product of the force and the	
---	-----	--	--	--